

AI-Powered E-Commerce Business Analysis

Excel Dashboard & AI-Assisted Business Intelligence Project

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Project Type: Data Analytics Portfolio Project

Tools & Technologies:

Microsoft Excel | PivotTables | PivotCharts | Excel Dashboard | AI-Assisted Analysis

Project Overview

This project presents an **AI-Powered E-Commerce Business Analysis** developed using Microsoft Excel and AI-assisted analysis techniques. The project uses transactional e-commerce data to evaluate overall business performance and identify meaningful patterns across categories, regions, months, products, and payment methods.

The analysis begins with data preparation and Excel-based calculations, followed by PivotTables and an interactive dashboard to summarize key performance indicators. AI-assisted analysis is then used to interpret the summarized results, identify important business patterns, and develop practical recommendations.

The objective is to transform raw transactional data into a structured **business intelligence solution** that can support data-driven decision-making.

Project at a Glance

Metric	Value
Total Orders	15,000
Units Sold	31,479
Total Revenue	₹59.60M
Total Profit	₹25.93M
Overall Profit Margin	43.50%

Analysis Areas

The project evaluates business performance across five major dimensions:

- Category Performance
- Regional Performance
- Monthly Performance

- Product Performance
- Payment Method Performance

Key Analytical Approach

Raw Transaction Data → Data Preparation → PivotTables → Dashboard → AI-Assisted Insights
→ Business Recommendations

Business Problem

The e-commerce dataset contains a large number of transactional records, but raw transaction-level data can make it difficult to understand overall business performance and identify important trends.

The project addresses the need to transform this raw data into meaningful business information by answering questions such as:

- Which categories generate the highest revenue and profit?
- Which regions contribute most to overall business performance?
- How does performance change across different months?
- Which products generate strong revenue and profitability?
- Which payment methods contribute most to revenue and profit?
- Where are potential opportunities for improving profitability and business performance?

Project Objectives

The primary objectives of this project are to:

1. **Analyze overall business performance** using revenue, profit, profit margin, orders, and units sold.
2. **Evaluate category performance** to identify the strongest and weakest categories based on revenue, profit, and profit margin.
3. **Analyze regional performance** to understand differences in revenue, profit, and profitability across regions.

4. **Identify monthly performance patterns** and determine periods of stronger or weaker business activity.
5. **Evaluate product performance** to identify high-revenue, high-profit, and high-margin products.
6. **Analyze payment methods** to understand their contribution to revenue and profit.
7. **Develop an interactive Excel dashboard** that presents important business KPIs and performance patterns in a concise visual format.
8. **Use AI-assisted analysis** to convert summarized analytical results into business insights and practical recommendations.

Dataset Overview

The project is based on an **e-commerce transactional dataset** containing **15,000 orders** and **31,479 units sold**.

The dataset was prepared and analyzed to evaluate business performance across different dimensions, including:

- Order date
- Region
- Category
- Product
- Payment method
- Revenue
- Profit
- Profit Margin

The dataset provides the foundation for calculating key performance indicators and creating summarized business views using Excel PivotTables.

Analysis Methodology

The analysis followed a structured workflow:

1. Data Preparation

The transactional dataset was reviewed and prepared for analysis. Relevant fields and calculated metrics were organized to support business performance analysis.

2. Excel-Based Analysis

Excel formulas were used to calculate and evaluate metrics such as **Revenue, Profit, and Profit Margin**.

3. PivotTable Analysis

PivotTables were created to summarize performance across:

- Category
- Region
- Order Date / Month
- Product
- Payment Method

4. Dashboard Development

PivotTables and PivotCharts were used to create an interactive **Excel business dashboard** presenting key KPIs and performance patterns.

5. AI-Assisted Analysis

The summarized results from the PivotTables were provided to an AI tool for structured interpretation. The AI analysis was used to identify:

- Key performance indicators
- Important patterns and comparisons
- Business insights
- Potential areas for improvement
- Practical business recommendations

6. Business Recommendations

The analytical findings were translated into actionable recommendations focused on improving profitability, managing growth, and supporting data-driven decision-making.

Analysis Workflow

Raw Dataset

↓

Data Preparation

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Excel Formulas & Calculations

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PivotTables & PivotCharts

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Interactive Dashboard

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AI-Assisted Insights

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Business Recommendations

Dashboard Overview

An interactive Excel dashboard was developed to provide a consolidated view of the e-commerce business performance. The dashboard brings together key performance indicators and visual summaries from the underlying PivotTable analysis.

The dashboard enables users to quickly evaluate:

- **Total Revenue**
- **Total Profit**
- **Profit Margin**
- **Total Orders**
- **Units Sold**

- Category Performance
- Regional Performance
- Monthly Performance
- Product Performance
- Payment Method Performance

The dashboard is designed to make the analysis easier to understand by presenting important business metrics and trends in a visual and structured format.

Dashboard Purpose

The primary purpose of the dashboard is to transform summarized transactional data into an easy-to-interpret business intelligence view. Instead of reviewing individual transaction records or multiple PivotTables separately, users can use the dashboard to identify important performance patterns and areas requiring further investigation.

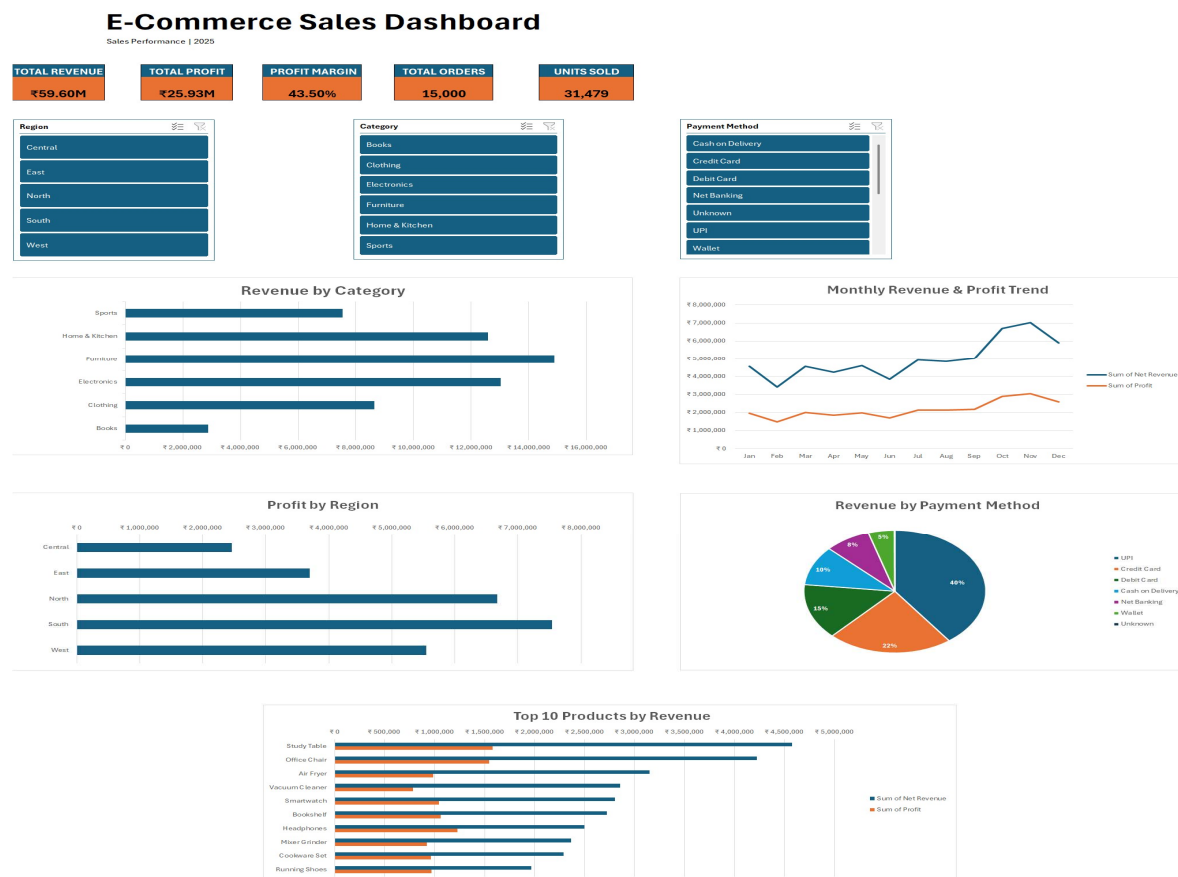


Figure 1: Interactive E-Commerce Business Performance Dashboard developed in Microsoft Excel

Category Performance Analysis

The category analysis compares revenue, profit, and profit margin across the six product categories.

Key Finding: Furniture generates the highest revenue, while Electronics generates the highest absolute profit.

Furniture contributes **₹14.90M in revenue** with a **38.13% profit margin**. Electronics generates **₹13.03M in revenue** but achieves a higher **₹5.81M profit** with a **44.63% margin**.

This demonstrates that the category generating the highest revenue is not necessarily the category generating the highest profit.

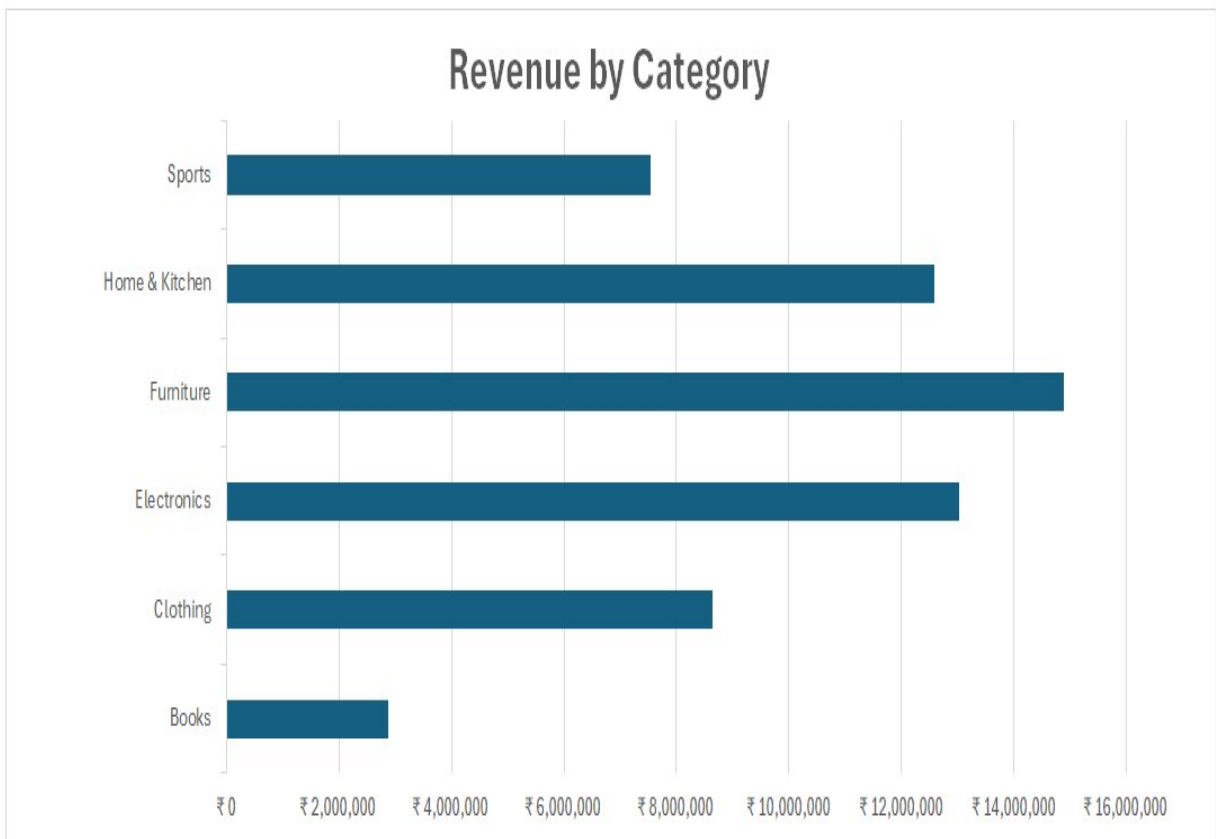


Figure 2: Revenue by Category

Business Insight

Furniture is an important revenue driver, but its lower margin indicates an opportunity to review pricing, costs, and discount strategies. Electronics demonstrates a stronger balance between sales scale and profitability.

Regional Performance Analysis

The regional analysis evaluates revenue, profit, and profit margin across Central, East, North, South, and West.

Key Finding: South is the strongest region by revenue and absolute profit.

South generates approximately **₹17.34M in revenue** and **₹7.54M in profit**, making it the largest regional contributor. At the same time, regional profit margins remain relatively close, ranging from approximately **43.22% to 44.19%**.



Figure 3: Profit by Region

Business Insight

The difference in absolute profit between regions is primarily associated with differences in revenue scale rather than major differences in profitability. South's performance should therefore be protected while lower-scale regions can be evaluated for opportunities to increase revenue.

Monthly Performance Analysis

The monthly analysis evaluates revenue, profit, and profit margin across the 12 months of the year to identify changes in business performance over time.

Key Finding: Business performance is concentrated toward the end of the year, with Q4 contributing the largest share of annual revenue.

The analysis shows that the business experiences stronger performance during the latter part of the year. Q4 represents the strongest quarterly period, while the final four months account for a substantial share of annual revenue.

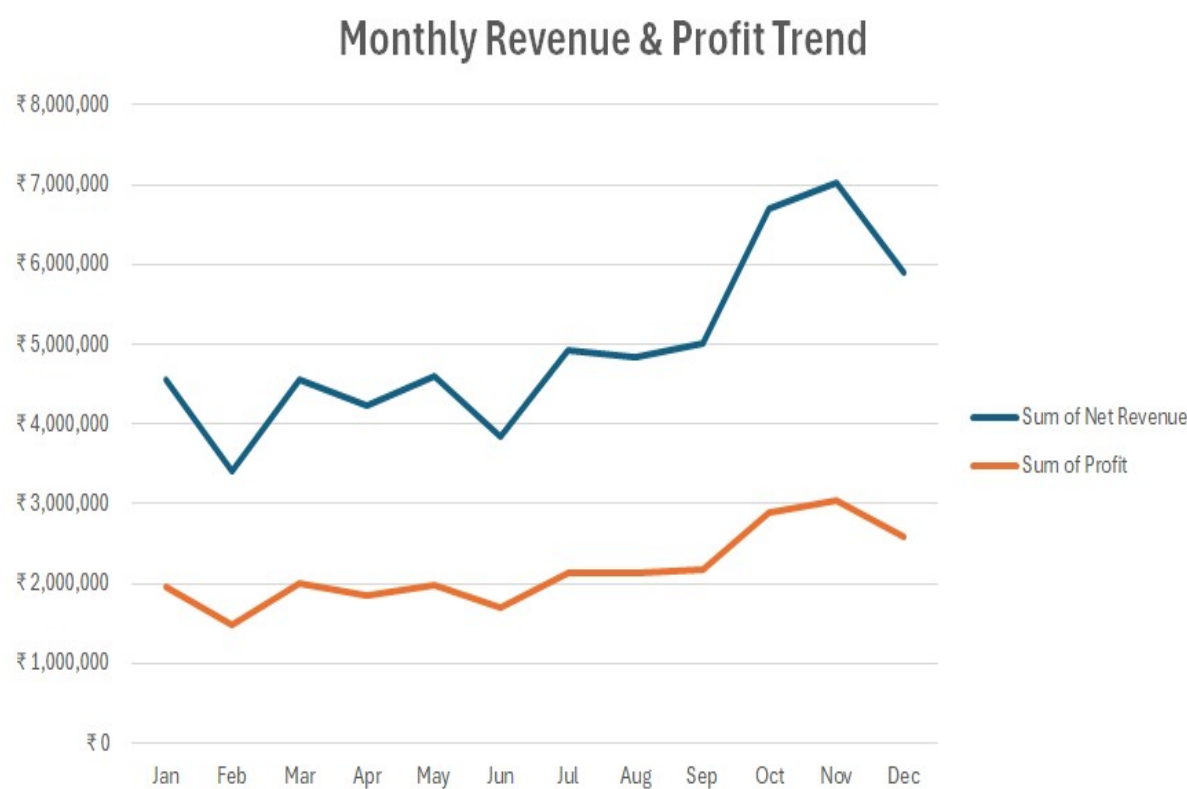


Figure 4: Monthly Revenue and Profit Performance

Business Insight

The concentration of revenue toward year-end suggests that inventory, staffing, fulfillment capacity, and marketing activities should be planned in advance of the stronger demand period.

Product Performance Analysis

The product analysis evaluates the Top 10 products based on revenue, profit, and profit margin.

Key Finding: Higher product revenue does not always translate into higher profitability.

Study Table is the strongest revenue and absolute-profit contributor among the Top 10 products, generating approximately **₹4.57M in revenue** and **₹1.58M in profit**. However, its profit margin is **34.51%**.

In comparison, Headphones generates approximately **₹2.50M in revenue** but achieves a much higher **49.18% profit margin**.

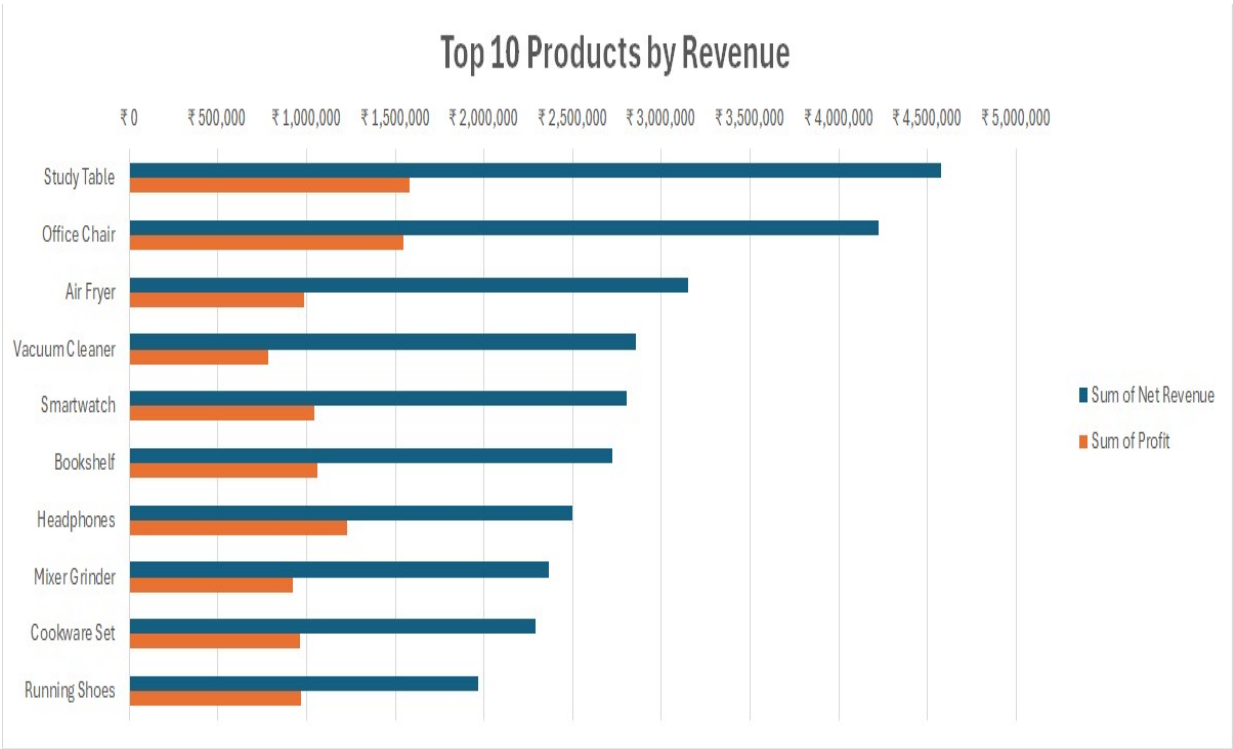


Figure 5: Top 10 Product Performance — Revenue, Profit

Business Insight

The comparison demonstrates the importance of evaluating both **sales volume and margin efficiency**. High-revenue products can generate substantial absolute profit while still having opportunities for margin improvement.

High-margin products such as **Running Shoes and Headphones** also represent potential opportunities for increasing sales volume while maintaining healthy profitability.

Payment Method Performance Analysis

The payment method analysis evaluates how different payment channels contribute to overall revenue, profit, and profit margin.

Key Finding: UPI is the dominant payment method and the largest contributor to both revenue and profit.

UPI generates approximately **₹23.74M in revenue** and **₹10.37M in profit**, making it the largest contributor among the payment methods. Credit Card is the second-largest contributor, generating approximately **₹13.28M in revenue** and **₹5.80M in profit**.

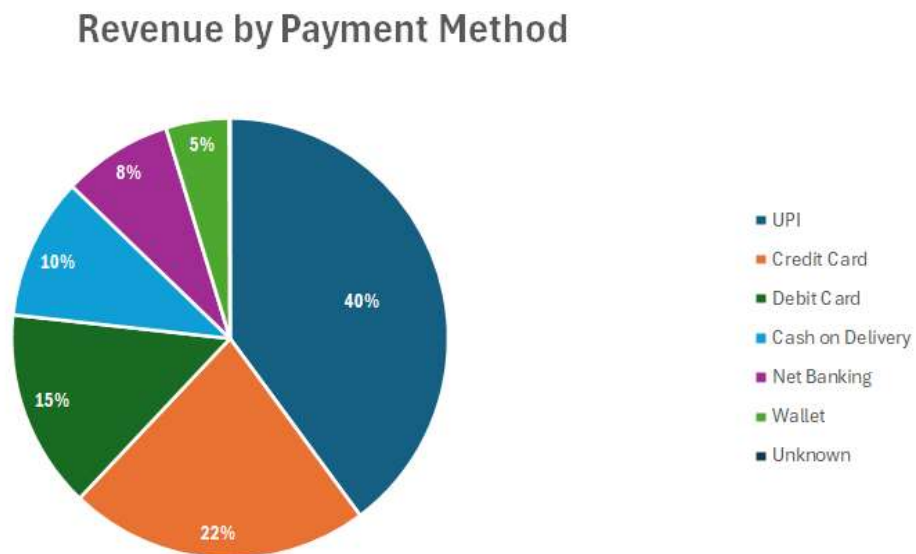


Figure 6: Revenue by Payment Method

Business Insight

The analysis shows that payment-method performance is primarily influenced by transaction volume, while profit margins across the established payment methods remain relatively consistent.

UPI and Credit Card are therefore important channels for maintaining transaction volume and overall business performance.

Data Quality Observation

The analysis also identified an **Unknown** payment-method classification in the original data. The Unknown records were retained in the analysis and their revenue and profit values were verified against the PivotTable results.

AI-Assisted Business Insights

After completing the Excel-based analysis and dashboard, AI was used as an analytical support tool to interpret the summarized results. The AI-assisted analysis focused on identifying significant performance patterns, comparing business dimensions, and converting data observations into actionable business insights.

The AI analysis covered **category, regional, monthly, product, and payment method performance**.

Analysis Area	Key Insight	Business Meaning
Category	Furniture generates the highest revenue, while Electronics generates the highest absolute profit.	Revenue leadership does not necessarily translate into profit leadership.
Region	South is the strongest region by revenue and absolute profit.	South is currently the largest regional contributor to business performance.
Monthly	Business performance is stronger toward the end of the year.	Year-end demand requires appropriate operational and inventory planning.
Product	Higher product revenue does not always mean higher profitability.	Both sales volume and profit margin should be considered when evaluating products.
High-Margin Products	Running Shoes and Headphones have strong profit margins among the Top 10 products.	These products may provide opportunities for profitable sales growth.
Payment Method	UPI is the dominant payment method and the largest contributor to revenue and profit.	Maintaining reliable UPI availability is important for transaction performance.

Role of AI in the Project

AI was **not used as a replacement for the Excel analysis**. The underlying calculations, PivotTables, dashboard, and summarized metrics were prepared in Excel first.

AI was then used to:

- Interpret the summarized analytical results
- Identify meaningful comparisons and patterns
- Distinguish data facts from business implications
- Generate potential business recommendations
- Support the conversion of analytical results into a business-oriented narrative

This approach combines **traditional Excel-based data analysis with AI-assisted interpretation**.

Business Recommendations

Based on the Excel analysis and AI-assisted interpretation, the following recommendations can support improved business performance and profitability.

1. Improve Furniture Profitability

Furniture is the highest-revenue category but operates at a lower profit margin than several other categories. Review pricing, discount levels, product costs, and other cost drivers to identify opportunities to improve profitability while protecting sales volume.

2. Protect South's Regional Performance

South is the strongest region by revenue and absolute profit. Maintain its current performance while examining successful practices that could provide useful benchmarks for lower-scale regions.

3. Prepare for Year-End Demand

The analysis indicates stronger business performance toward the end of the year. Inventory, staffing, fulfillment, and marketing capacity should be planned ahead of this period to handle increased demand effectively.

4. Review Low-Margin Products

Products such as **Vacuum Cleaner** and **Air Fryer** generate meaningful revenue but have comparatively lower profit margins. Review their pricing, discounts, product costs, and fulfillment expenses to identify margin-improvement opportunities.

5. Scale High-Margin Products

Running Shoes and **Headphones** demonstrate strong profit margins among the Top 10 products. Promotions, bundles, cross-selling, and improved product visibility could be tested to increase their sales volume while protecting margins.

6. Maintain Strong Payment Channels

UPI and Credit Card are major contributors to payment-method revenue and profit. Reliable processing and strong checkout visibility should be maintained for these channels while continuing to support alternative payment options.

7. Maintain Margin Discipline During Growth

The analysis shows relatively stable profitability across several business dimensions. Future growth initiatives should therefore focus on increasing revenue while continuing to monitor pricing, costs, discounts, and product-level margins.

Recommendation Summary

Area	Priority Action
Furniture	Improve margin efficiency
South Region	Protect current performance
Year-End	Prepare capacity and inventory
Low-Margin Products	Review costs and pricing
High-Margin Products	Increase sales volume
Payment Methods	Maintain reliable key channels
Overall Growth	Protect profitability while scaling

Conclusion

This project transformed 15,000 e-commerce orders and 31,479 units sold into a structured business analysis using Microsoft Excel and AI-assisted interpretation.

The analysis evaluated overall business performance and examined revenue, profit, and profitability across categories, regions, months, products, and payment methods. Excel PivotTables and dashboard visualizations were used to summarize the data, while AI-assisted analysis helped interpret the results and develop practical business recommendations.

The findings demonstrate that revenue scale and profitability efficiency do not always move together. Furniture leads in revenue, while Electronics generates the highest absolute profit. South is the strongest region by scale, business performance is stronger toward the end of the year, and high-margin products such as Running Shoes and Headphones present potential growth opportunities. UPI is also the dominant payment method.

Overall, the project demonstrates how Excel-based data analysis combined with AI-assisted interpretation can transform transactional data into a practical business intelligence solution that supports data-driven decision-making.